

## WTW 158 The 55 Integration Challenge

1.  $\int \frac{x}{\sqrt{1-x^2}} dx$
2.  $\int (2x+1)(x^2+x)^4 dx$
3.  $\int \frac{\sec^2 x}{1+\tan x} dx$
4.  $\int \frac{2x}{(1+x^2)^3} dx$
5.  $\int \frac{x}{\sqrt{1-4x^2}} dx$
6.  $\int \frac{4}{\sqrt{1-x^2}} dx$
7.  $\int 5^{2x} dx$
8.  $\int \frac{1}{1+x^2} dx$
9.  $\int \frac{1}{x \ln x} dx$
10.  $\int x\sqrt{x^2-3} dx$
11.  $\int \frac{e^{3x}}{\sqrt[3]{1-e^{3x}}} dx$
12.  $\int \sqrt{2x+1} dx$
13.  $\int \sin^3 2\theta \cos 2\theta d\theta$
14.  $\int \cos x \sin x e^{\cos^2 x} dx$
15.  $\int \sin^3 3\theta \cos 3\theta d\theta$
16.  $\int \tan 2x dx$
17.  $\int \frac{\cos^3(\ln x) \sin(\ln x)}{x} dx$
18.  $\int \sec^2 x \tan^2 x dx$
19.  $\int x \sin x^2 e^{\cos x^2} dx$
20.  $\int \frac{1+\sin x}{x-\cos x} dx$
21.  $\int \sec^2 x e^{\tan x} dx$
22.  $\int 4x^3 \sqrt{6x^4-1} dx$
23.  $\int \frac{e^{\sqrt{t}}}{\sqrt{t}} dt$
24.  $\int \sin^4 \theta \cos \theta d\theta$
25.  $\int (3x-2)^4 dx$
26.  $\int \frac{2t-1}{t^2-t+2} dt$
27.  $\int \frac{1}{\sqrt{t}(1-2\sqrt{t})} dt$
28.  $\int \frac{3}{\sqrt{1-t^2}} dt$
29.  $\int t \sin(t^2) dt$

$$30. \int \sec 3t \tan 3t dt$$

$$31. \int \sin x e^{\cos x} dx$$

$$32. \int (-2x + 5)^{\frac{3}{2}} dx$$

$$33. \int \frac{2}{(t-9)^2} dt$$

$$34. \int \frac{t^2-3}{-t^3+9t+1} dt$$

$$35. \int \frac{e^x}{1+e^x} dx$$

$$36. \int \frac{x^2}{x-1} dx$$

$$37. \int x \cos(\pi x^2) dx$$

$$38. \int \cos ec(\pi x) \cot(\pi x) dx$$

$$39. \int \frac{\sin x}{\sqrt{\cos x}} dx$$

$$40. \int \frac{2}{e^{-x}+1} dx$$

$$41. \int \frac{e^{\frac{1}{t}}}{t^2} dt$$

$$42. \int \sin^2 t \cos t dt$$

$$43. \int (1 + e^x)^2 dx$$

$$44. \int \frac{2^t}{1+2^t} dt$$

$$45. \int 5x \sin(x^2 - 1) dx$$

$$46. \int t^{\frac{1}{3}} (1 - t) dt$$

$$47. \int (1 - bx)^{-1} dx$$

$$48. \int \frac{\ln(x-5)}{x-5} dx$$

$$49. \int \frac{\sin x}{1 - \sin^2 x} dx$$

$$50. \int \frac{x}{1+4x^2} dx$$

$$51. \int \frac{5x+2}{\sqrt{x}} dx$$

$$52. \int \frac{x^3}{\sqrt{2x^4+3}} dx$$

$$53. \int \frac{e^{\frac{1}{x}}}{x^2} dx$$

$$54. \int \frac{e^x}{\sqrt{1-e^{2x}}} dx$$

$$55. \int \frac{1}{(at+b)^6} dt$$